

KUNAL ROY CHOUDHURY

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EXPERIENCE

Lead Software Engineer

July 2026 – Present

FinStocks AI

Bengaluru, Karnataka

- Architected a natural-language-to-strategy backtesting engine, leading a 3-engineer team; constrained LLM output to a typed **Pydantic IR** over **185 indicators** / **148 patterns**, with deterministic compilation and static validation.
- Accelerated backtesting with runtime **Numba-JIT kernels** and whole-universe fused execution over struct-of-arrays **float32 Parquet**; liveness analysis loads only strategy-referenced columns.
- Unified backtest and live execution through the same IR/compiler; implemented **tick coalescing** for bounded evaluation latency and **Redis Lua** for atomic capital allocation across workers.
- Built an **LLM chart terminal** with **85 typed operations** and symbolic price references; enforced level provenance through a 1.4k-line validator, preventing fabricated chart levels.
- Engineered transactional push on **Postgres** using SKIP LOCKED, visibility leases and idempotent partial indexes; optimized market-data queries from **13.6s to 48–135ms** and **5.9s to 782ms**, streaming bars via **Redis Pub/Sub** to a TradingView datafeed.

Software Engineer

March 2026 – June 2026

CollectEdge

Bengaluru, India

- Scaled to **50K+ daily messages** across 5+ AWS SQS queues with **Rust** microservices, adding backpressure, circuit breakers, and idempotent processing to eliminate duplicate-delivery failures.
- Built **CredFix**'s **sub-100ms** credit scoring with LRU caching, batching, and 18 parallel rule workers.

Software Developer

April 2025 – February 2026

Indian Kanoon

Bengaluru, India

- Sole engineer on **Prism**, a legal research product launched into a **1.5M-user** platform, built on **Django**, **PostgreSQL**, and **Redis** with LLM workflows spanning 10+ tools.
- Achieved **40% lower latency** and **92% query accuracy** in a **RAG** retrieval stack using PostgreSQL full-text search, query expansion, caching, and index tuning.

Software Engineer Intern

September 2024 – April 2025

Chargebee

Chennai, India

- Migrated **500K+ subscriptions** with zero customer-facing downtime using a migration framework built with **JOJO** schedulers and **Vue.js** validation tooling.

Software Engineer Intern

January 2024 – September 2024

AiDash

Bengaluru, India

- Contributed to a **60% faster** release cycle by extracting services from a **2M+ LOC monolith** using **Spring Boot** and hexagonal architecture; optimized SQL pagination over 100K+ record tables.

PROJECTS

Codebase Indexer | *Rust, Tree-sitter, Tantivy, ONNX, HNSW, RRF Fusion*

[code](#) | [article](#)

- Indexed **50K files** in **21 seconds** with a local **Rust** hybrid search engine combining Tree-sitter chunking, Tantivy BM25, and HNSW retrieval via RRF; used **INT8 ONNX** models for local inference.

Gostman | *Go, React, Wails, REST, GraphQL, WebSockets*

[live](#) | [code](#)

- Achieved **<1s cold launch** and **90% lower memory** than Electron equivalents with a privacy-first desktop API client featuring local-only storage and request chaining across REST, GraphQL, and WebSocket protocols.

TECHNICAL SKILLS

Languages: Go, Rust, Python, Java, TypeScript, SQL

Systems & Performance: concurrency, cgo/FFI, profiling, caching, batching, backpressure, circuit breakers

AI / LLM: LangGraph, RAG, tool calling, vector search (HNSW), ONNX, Anthropic & OpenAI APIs

Backend: Django, FastAPI, Spring Boot, gRPC, REST, GraphQL, WebSockets, WebRTC, Celery

Data & Infra: PostgreSQL, Redis, Elasticsearch, AWS, GCP, Docker, Kubernetes, CI/CD

Frontend: React, React Native (Expo), Vue.js

EDUCATION

SRM Institute of Science and Technology

B.Tech, Computer Science and Engineering — CGPA: 9.24 / 10

Kattankulathur, India

August 2020 – September 2024